SCIKEY DATABASE — DBA RUNBOOK

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1. Overview & Scope

This runbook covers routine DBA maintenance for the SCIKEY Oracle 19c database. All operations assume you are connected to the pluggable database.

ALTER SESSION SET CONTAINER = pdb;

Schema Summary

Table	Type	Partitioning	~Rows
keyword	Partitioned	RANGE on keyword_text (A–Z + other)	868,000
document	Composite partitioned	LIST(discipline) × HASH(hal_doc_id)	750,000
doc_keyword	Reference partitioned	Inherits from document	Large
doc_author	Reference partitioned	Inherits from document	Large
author_organism	Reference partitioned	Inherits from doc_author	Large
author	Unpartitioned (dimension)	—	Medium
organism	Unpartitioned (dimension)	—	Small

Tablespaces

Tablespace	Purpose
scikey_data	All table and LOB segments
scikey_index	All index segments

---

2. Index Rebuilds

When to Rebuild

Rebuild indexes when:

- `INDEX_STATS` shows DELETED_LEAF_ROWS / LEAF_ROWS > 20%
- After a large bulk delete or partition purge
- After `ALTER TABLE ... MOVE` (which invalidates all local indexes)
- Optimizer is ignoring an index despite low selectivity

Check Index Health First

```
-- Check for unusable indexes
SELECT index_name, partition_name, status
FROM   dba_ind_partitions
WHERE  table_owner = 'SCIKEY'
  AND  status = 'UNUSABLE'
ORDER  BY index_name, partition_name;

-- Check for bloat (run ANALYZE first, then query)
ANALYZE INDEX scikey.idx_keyword_text VALIDATE STRUCTURE;

SELECT name, height, lf_rows, del_lf_rows,
       ROUND(del_lf_rows / NULLIF(lf_rows, 0) * 100, 1) AS pct_deleted
FROM   index_stats;
```

Rebuild All Local Indexes on a Single Partition

Use this after moving or purging a specific partition.

```
-- Example: rebuild all local indexes for keyword partition p_kw_s
ALTER INDEX scikey.idx_keyword_text      REBUILD PARTITION p_kw_s  TABLESPACE
scikey_index ONLINE;
ALTER INDEX scikey.idx_keyword_text_upper REBUILD PARTITION p_kw_s TABLESPACE
scikey_index ONLINE;
```

Rebuild a Specific Index (Full, All Partitions)

```
-- LOCAL index: rebuilds all partitions in one statement
ALTER INDEX scikey.idx_keyword_text       REBUILD TABLESPACE scikey_index ONLINE;
ALTER INDEX scikey.idx_keyword_text_upper REBUILD TABLESPACE scikey_index ONLINE;
ALTER INDEX scikey.idx_document_hal_id    REBUILD TABLESPACE scikey_index ONLINE;
ALTER INDEX scikey.idx_document_doi       REBUILD TABLESPACE scikey_index ONLINE;
ALTER INDEX scikey.idx_doc_keyword_document REBUILD TABLESPACE scikey_index
ONLINE;
ALTER INDEX scikey.idx_dk_keyword         REBUILD TABLESPACE scikey_index ONLINE;
```

```
ALTER INDEX scikey.idx_doc_author_document REBUILD TABLESPACE scikey_index
ONLINE;
ALTER INDEX scikey.idx_da_author           REBUILD TABLESPACE scikey_index ONLINE;
ALTER INDEX scikey.idx_ao_organism         REBUILD TABLESPACE scikey_index ONLINE;
ALTER INDEX scikey.idx_ao_document_author  REBUILD TABLESPACE scikey_index
ONLINE;

-- GLOBAL (unpartitioned) indexes on author dimension table
ALTER INDEX scikey.idx_author_name         REBUILD TABLESPACE scikey_index ONLINE;
ALTER INDEX scikey.idx_author_lastname_upper REBUILD TABLESPACE scikey_index
ONLINE;
```

Rebuild Bitmap Indexes

WARNING: Bitmap indexes cannot be rebuilt ONLINE. Schedule during a maintenance window with no concurrent DML.

```
-- Drop and recreate (safest approach for bitmaps)
DROP INDEX scikey.bidx_document_type;
CREATE BITMAP INDEX scikey.bidx_document_type
   ON scikey.document(document_type)
   LOCAL TABLESPACE scikey_index;

DROP INDEX scikey.bidx_document_discipline;
CREATE BITMAP INDEX scikey.bidx_document_discipline
   ON scikey.document(discipline)
   LOCAL TABLESPACE scikey_index;

DROP INDEX scikey.bidx_doc_author_quality;
CREATE BITMAP INDEX scikey.bidx_doc_author_quality
   ON scikey.doc_author(quality)
   LOCAL TABLESPACE scikey_index;
```

Mark All Indexes Usable After Move

If you ran ALTER TABLE ... MOVE on the document table (or any partitioned table), all local indexes will be marked UNUSABLE. Rebuild them:

```
-- Generate rebuild statements for all UNUSABLE partitioned indexes
SELECT 'ALTER INDEX scikey.' || i.index_name
   || ' REBUILD PARTITION ' || p.partition_name
   || ' TABLESPACE scikey_index ONLINE;' AS stmt
FROM   dba_indexes i
```

```
JOIN   dba_ind_partitions p ON p.index_name = i.index_name
WHERE  i.table_owner = 'SCIKEY'
  AND  p.status = 'UNUSABLE'
ORDER  BY i.index_name, p.partition_name;
```

Copy and execute the output.

---

3. Statistics Updates

Oracle's cost-based optimizer relies on accurate statistics. SCIKEY uses partitioned tables with skewed distributions, so partition-level stats matter.

Recommended Defaults for SCIKEY

Parameter	Value	Reason
estimate_percent	AUTO_SAMPLE	Oracle determines sample size automatically
method_opt	FOR ALL COLUMNS SIZE AUTO	Auto-histograms on skewed cols
degree	4	Parallelism — adjust to available CPUs
cascade	TRUE	Also gathers index statistics
granularity	ALL	Global + partition + subpartition stats

Gather Stats on All SCIKEY Tables

Run after a bulk data load or at the weekly maintenance window.

```
BEGIN
  DBMS_STATS.GATHER_SCHEMA_STATS(
    ownname          => 'SCIKEY',
    estimate_percent => DBMS_STATS.AUTO_SAMPLE_SIZE,
    method_opt       => 'FOR ALL COLUMNS SIZE AUTO',
    degree           => 4,
    cascade          => TRUE,
    granularity      => 'ALL'
  );
END;
```

/

Gather Stats on a Single Table

Use this after targeted bulk inserts (e.g., adding new HAL documents).

```
BEGIN
  DBMS_STATS.GATHER_TABLE_STATS(
    ownname          => 'SCIKEY',
    tabname          => 'DOCUMENT',
    estimate_percent => DBMS_STATS.AUTO_SAMPLE_SIZE,
    method_opt       => 'FOR ALL COLUMNS SIZE AUTO',
    degree           => 4,
    cascade          => TRUE,
    granularity      => 'ALL'
  );
END;
/
```

Replace `'DOCUMENT'` with `'KEYWORD'`, `'AUTHOR'`, `'ORGANISM'`, `'DOC_KEYWORD'`, `'DOC_AUTHOR'`, or `'AUTHOR_ORGANISM'` as needed.

Gather Stats on a Single Partition

Use after loading data into one discipline partition of DOCUMENT.

```
BEGIN
  DBMS_STATS.GATHER_TABLE_STATS(
    ownname          => 'SCIKEY',
    tabname          => 'DOCUMENT',
    partname         => 'P_CS',          -- e.g., Computer Science partition
    estimate_percent => DBMS_STATS.AUTO_SAMPLE_SIZE,
    method_opt       => 'FOR ALL COLUMNS SIZE AUTO',
    degree           => 4,
    cascade          => TRUE,
    granularity      => 'PARTITION'
  );
END;
/
```

Valid partition names for DOCUMENT: P_UNKNOWN, P_CS, P_CHEMENG, P_MARKETING, P_POLISCI, P_CIVIL

Lock Statistics (Prevent Auto-Updates)

If a table has stable, well-tuned stats that the auto-job keeps overwriting:

```
DBMS_STATS.LOCK_TABLE_STATS(ownname => 'SCIKEY', tabname => 'ORGANISM');
-- To unlock:
DBMS_STATS.UNLOCK_TABLE_STATS(ownname => 'SCIKEY', tabname => 'ORGANISM');
```

Check Statistics Freshness

```
SELECT table_name,
       num_rows,
       last_analyzed,
       ROUND(SYSDATE - last_analyzed, 1) AS days_since_analyzed
FROM   dba_tables
WHERE  owner = 'SCIKEY'
ORDER  BY last_analyzed NULLS FIRST;
```

Rule of thumb: Any table with days_since_analyzed > 7 after active loading should be re-analyzed.

Check for Missing or Stale Column Histograms

```
SELECT table_name, column_name, histogram, num_distinct, num_nulls, last_analyzed
FROM   dba_tab_col_statistics
WHERE  owner = 'SCIKEY'
  AND  column_name IN ('DISCIPLINE', 'DOCUMENT_TYPE', 'QUALITY', 'KEYWORD_TEXT')
ORDER  BY table_name, column_name;
```

Histogram should be FREQUENCY or HEIGHT BALANCED for low-cardinality columns (discipline, document_type, quality). If it shows NONE, re-gather with SIZE AUTO.

---

4. Space Management

Check Tablespace Usage

```
SELECT df.tablespace_name,
       ROUND(df.total_mb, 0)                    AS total_mb,
       ROUND(df.total_mb - fs.free_mb, 0)          AS used_mb,
       ROUND(fs.free_mb, 0)                     AS free_mb,
       ROUND((df.total_mb - fs.free_mb) / df.total_mb * 100, 1) AS pct_used
```

```
FROM (
   SELECT tablespace_name, SUM(bytes) / 1048576 AS total_mb
   FROM   dba_data_files GROUP BY tablespace_name
) df
JOIN (
   SELECT tablespace_name, SUM(bytes) / 1048576 AS free_mb
   FROM   dba_free_space GROUP BY tablespace_name
) fs ON fs.tablespace_name = df.tablespace_name
WHERE  df.tablespace_name IN ('SCIKEY_DATA', 'SCIKEY_INDEX')
ORDER  BY df.tablespace_name;
```

Alert if either tablespace exceeds 80% used.

Check Segment Sizes by Table

```
SELECT segment_name, segment_type, partition_name,
      ROUND(SUM(bytes) / 1048576, 1) AS size_mb
FROM   dba_segments
WHERE  owner = 'SCIKEY'
GROUP  BY segment_name, segment_type, partition_name
ORDER  BY SUM(bytes) DESC
FETCH  FIRST 30 ROWS ONLY;
```

Check LOB Segment Sizes

The DOCUMENT table stores abstract, domain_codes, and url_primary as CLOBs. Check their segments separately.

```
SELECT l.table_name, l.column_name, l.segment_name,
      ROUND(SUM(s.bytes) / 1048576, 1) AS lob_mb
FROM   dba_lobs l
JOIN   dba_segments s ON s.segment_name = l.segment_name AND s.owner = l.owner
WHERE  l.owner = 'SCIKEY'
GROUP  BY l.table_name, l.column_name, l.segment_name
ORDER  BY SUM(s.bytes) DESC;
```

Reclaim Space After Large Deletes

After deleting rows, Oracle does not automatically return space to the tablespace. Use segment shrink or move to reclaim it.

Option A — Shrink in place (online, preferred):

```
-- Enable row movement first
ALTER TABLE scikey.document ENABLE ROW MOVEMENT;

-- Shrink (COMPACT first to avoid blocking, then full shrink)
ALTER TABLE scikey.document SHRINK SPACE COMPACT;
ALTER TABLE scikey.document SHRINK SPACE CASCADE;  -- also shrinks indexes

-- Disable row movement when done
ALTER TABLE scikey.document DISABLE ROW MOVEMENT;
```

Note: SHRINK CASCADE does not work on LOB segments. Shrink LOBs separately if needed.

Option B — Move partition (offline, more thorough):

```
-- Moves a single partition, compacts it, resets HWM
-- Indexes on this partition become UNUSABLE — rebuild afterwards
ALTER TABLE scikey.document MOVE PARTITION p_cs TABLESPACE scikey_data;

-- Then rebuild local indexes for that partition (see Section 2)
```

Add Space to a Tablespace

```
-- Add a datafile
ALTER TABLESPACE scikey_data ADD DATAFILE '/path/to/scikey_data02.dbf' SIZE 10G
AUTOEXTEND ON NEXT 512M MAXSIZE 50G;
ALTER TABLESPACE scikey_index ADD DATAFILE '/path/to/scikey_index02.dbf' SIZE 5G
AUTOEXTEND ON NEXT 256M MAXSIZE 20G;
```

WARNING: Confirm the datafile path with your storage team before running. Placing files in the wrong location can cause recovery complications.

Resize an Existing Datafile

```
-- Increase size of an existing datafile
ALTER DATABASE DATAFILE '/path/to/scikey_data01.dbf' RESIZE 20G;
```

---

5. Partition Maintenance

Add a New Discipline Partition to DOCUMENT

If a new discipline is ingested from HAL that is not covered by existing LIST values, add a partition before loading.

```
ALTER TABLE scikey.document
   ADD PARTITION p_newdiscipline
   VALUES ('New Discipline Name')
   SUBPARTITIONS 2;
```

The reference-partitioned tables (doc_keyword, doc_author, author_organism) will automatically receive matching partitions — no action needed on them.

Add a New Keyword Partition

The keyword table uses RANGE partitioning on keyword_text. To split the catch-all p_kw_other partition (e.g., to carve out a dedicated partition for a new character range):

```
ALTER TABLE scikey.keyword
   SPLIT PARTITION p_kw_other
   AT ('NewBoundary')
   INTO (
      PARTITION p_kw_newrange TABLESPACE scikey_data,
      PARTITION p_kw_other    TABLESPACE scikey_data
   );
```

Rebuild the local indexes on both resulting partitions afterwards.

Drop an Empty Partition

```
-- Verify it is empty first
SELECT COUNT(*) FROM scikey.document PARTITION (p_marketing);

-- Then drop
ALTER TABLE scikey.document DROP PARTITION p_marketing UPDATE INDEXES;
```

UPDATE INDEXES keeps local indexes valid; without it they become UNUSABLE.

Check Partition Row Counts

```
-- Uses partition-level stats (fast, approximate)
SELECT table_name, partition_name, num_rows, last_analyzed
FROM   dba_tab_partitions
WHERE  table_owner = 'SCIKEY'
  AND  table_name  = 'DOCUMENT'
```

```
ORDER  BY partition_position;
```

Check Subpartition Distribution for DOCUMENT

```
SELECT partition_name, subpartition_name, num_rows
FROM   dba_tab_subpartitions
WHERE  table_owner = 'SCIKEY'
  AND  table_name  = 'DOCUMENT'
ORDER  BY partition_name, subpartition_name;
```

---

6. Health Check Queries

Run these during regular DBA checks or when investigating performance issues.

Invalid Objects

```
SELECT object_type, object_name, status
FROM   dba_objects
WHERE  owner  = 'SCIKEY'
  AND  status = 'INVALID'
ORDER  BY object_type, object_name;
```

No rows expected. If rows appear after DDL changes, recompile:

```
EXEC DBMS_UTILITY.COMPILE_SCHEMA(schema => 'SCIKEY', compile_all => FALSE);
```

Unusable Indexes

```
SELECT index_name, partition_name, subpartition_name, status
FROM   dba_ind_partitions
WHERE  index_owner = 'SCIKEY'
  AND  status      = 'UNUSABLE'
UNION ALL
SELECT index_name, NULL, NULL, status
FROM   dba_indexes
WHERE  owner  = 'SCIKEY'
  AND  status = 'UNUSABLE'
ORDER  BY 1, 2, 3;
```

Disabled Constraints

```
SELECT constraint_name, constraint_type, table_name, status
FROM   dba_constraints
WHERE  owner  = 'SCIKEY'
  AND  status = 'DISABLED'
ORDER  BY table_name, constraint_name;
```

Sequence Values

```
SELECT sequence_name, last_number, increment_by, max_value
FROM   dba_sequences
WHERE  sequence_owner = 'SCIKEY'
ORDER  BY sequence_name;
```

Watch for sequences approaching MAX_VALUE (default unlimited, but verify).

Long-Running Sessions

```
SELECT s.sid, s.serial#, s.username, s.status, s.sql_id,
       ROUND(l.time_remaining / 60, 1) AS mins_remaining,
       ROUND(l.elapsed_seconds / 60, 1) AS mins_elapsed,
       l.message
FROM   v$session_longops l
JOIN   v$session s ON s.sid = l.sid AND s.serial# = l.serial#
WHERE  l.time_remaining > 0
  AND  l.username = 'SCIKEY'
ORDER  BY l.time_remaining DESC;
```

Active Locks

```
SELECT s.sid, s.serial#, s.username, s.status,
       o.object_name, l.type, l.lmode, l.request,
       l.block
FROM   v$lock l
JOIN   v$session s ON s.sid = l.sid
JOIN   dba_objects o ON o.object_id = l.id1
WHERE  s.username = 'SCIKEY'
  AND  l.block    = 1
ORDER  BY l.block DESC;
```

AWR Snapshot Check (if licensed)

```
SELECT snap_id, begin_interval_time, end_interval_time
FROM   dba_hist_snapshot
```

```
WHERE  begin_interval_time > SYSDATE - 2
ORDER  BY snap_id DESC
FETCH  FIRST 10 ROWS ONLY;
```

---

7. Emergency Reference

Kill a Blocking Session

```
-- Identify blocker
SELECT blocking_session, sid, serial#, wait_class, seconds_in_wait
FROM   v$session
WHERE  blocking_session IS NOT NULL;

-- Kill it
ALTER SYSTEM KILL SESSION 'sid,serial#' IMMEDIATE;
```

Force-Rebuild a Single Unusable Index Partition

```
ALTER INDEX scikey.<index_name> REBUILD PARTITION <partition_name>
   TABLESPACE scikey_index ONLINE;
```

Quickly Re-enable All Unusable Indexes (script generator)

```
SELECT 'ALTER INDEX scikey.' || index_name
   || CASE WHEN partition_name IS NOT NULL
          THEN ' REBUILD PARTITION ' || partition_name
          ELSE ' REBUILD'
      END
   || ' TABLESPACE scikey_index ONLINE;'
FROM (
   SELECT index_name, partition_name
   FROM   dba_ind_partitions
   WHERE  index_owner = 'SCIKEY' AND status = 'UNUSABLE'
   UNION ALL
   SELECT index_name, NULL
   FROM   dba_indexes
   WHERE  owner = 'SCIKEY' AND status = 'UNUSABLE'
);
```

Rollback a Failed Stats Gather

Oracle retains previous stats. Restore from the most recent snapshot:

```
DECLARE
  v_ts TIMESTAMP;
BEGIN
  SELECT MAX(savetime) INTO v_ts
  FROM   dba_optstat_operations
  WHERE  operation = 'gather_table_stats'
    AND  target    LIKE 'SCIKEY.%';

  DBMS_STATS.RESTORE_TABLE_STATS(
    ownname   => 'SCIKEY',
    tabname   => 'DOCUMENT',
    as_of_timestamp => v_ts - INTERVAL '1' MINUTE
  );
END;
/
```

Partition Reference Card

DOCUMENT partitions (LIST on discipline)

Partition	Discipline Value	Subpartitions
P_UNKNOWN	NULL	16
P_CS	Computer Science	8
P_CHEMENG	Chemical Engineering	4
P_MARKETING	Marketing	2
P_POLISCI	Political Science	2
P_CIVIL	Civil Engineering	2

KEYWORD partitions (RANGE on keyword_text)

Partition	Range	~Rows
P_KW_A	< 'B'	61,393
P_KW_B	< 'C'	29,092

P_KW_C	< 'D'	87,884
P_KW_D	< 'E'	45,564
P_KW_E	< 'F'	43,669
P_KW_F	< 'G'	34,368
P_KW_G	< 'H'	27,359
P_KW_H	< 'I'	27,654
P_KW_I	< 'J'	38,895
P_KW_J	< 'K'	6,230
P_KW_K	< 'L'	5,680
P_KW_L	< 'M'	34,994
P_KW_M	< 'N'	68,914
P_KW_N	< 'O'	24,170
P_KW_O	< 'P'	19,493
P_KW_P	< 'Q'	77,618
P_KW_Q	< 'R'	5,016
P_KW_R	< 'S'	44,646
P_KW_S	< 'T'	81,533
P_KW_T	< 'U'	44,056
P_KW_U	< 'V'	9,004
P_KW_V	< 'W'	17,025
P_KW_W	< 'X'	9,461
P_KW_XYZ	< 'a'	4,800
P_KW_OTHER	< MAXVALUE	20,035